Quiz 02 Weeks 03 to 06

Teaching Week 06 | Sample questions with the answers marked, for study.

Name _____

Banner ID _____ **Dalhousie email** _____

Circle one letter for each question. Each question has exactly one correct answer. Course notes and your own notes are permitted. Generative AI services and communication with other people are not permitted.

1. Two data sets have the same mean, variance and correlation. What should you do before calling them alike?

A. Plot both data sets
B. Report only the correlation
C. Remove the largest value
D. Compute the variance again
E. Nothing, equal summaries mean equal data

2. In a box plot, what does the line inside the box mark?

A. The mean
B. The largest value
C. The median
D. The most frequent value
E. The number of outliers

3. Seven play records match a game, one play record has no game, and one game has no play record. How many rows does the left join, plays on the left, give?

A. 7 **B. 8** C. 9 D. 15 E. 16

4. A report renders cleanly. What does that establish?

A. The sample represents the population
B. The cleaning choices were justified
C. The conclusions follow from the charts
D. Its source, data and software produced that document
E. The code gives the same result on any data

5. A test set has 100 cases, 10 of them positive. A model that always predicts negative has which error rate?

A. 0 **B. 0.1** C. 0.5 D. 0.9 E. 1

6. Which formula gives the precision for the positive class?

A. $TP/(TP + FN)$
B. $(TP + TN)/n$
C. $FP/(FP + TN)$
D. $TN/(TN + FP)$
E. $TP/(TP + FP)$

7. Five test predictions miss by 1, 1, 1, 1 and 6 units. Which is true?

A. MAE is 2 and RMSE is about 2.83
B. MAE and RMSE are both 2
C. MAE is 2 and RMSE is 8
D. RMSE is smaller than MAE
E. MAE is 10, the sum of the misses

8. Why are the predictors standardized before kNN?

A. kNN can only read values between 0 and 1
B. Standardizing removes outliers from the training set
C. A wide-range predictor would dominate the distance
D. It makes the classes equally frequent
E. It chooses the best value of K

9. Compared with $K = 15$, kNN with $K = 1$

A. assumes a straight-line boundary
B. has a higher training error
C. needs the predictors to be Normal
D. ignores the nearest case
E. follows noise in the training data and may overfit

10. In the algae model, `size` has baseline `large` and `sizesmall` has weight 9.75. What does that mean?

A. Every river is predicted 9.75 higher
B. Small rivers are 9.75 times more frequent
C. 9.75% of the rivers are small
D. A small river is predicted 9.75 higher than a large one, all else fixed
E. Large rivers are predicted 9.75 higher than small ones

SAMPLE QUESTIONS